

Task 5 – Risk of Bias Assessment

JBI Critical Appraisal of Two Experimental Neuroscience Articles

Prepared for: Task 5

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Assessment framework: Joanna Briggs Institute (JBI) critical appraisal approach

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1. Scope and methodological note

Both articles are laboratory-based experimental animal/in-vitro neuroscience studies rather than human observational studies. JBI provides design-specific critical appraisal tools, and its current tool library includes a quasi-experimental studies tool. Because the two papers do not fit human cohort or cross-sectional designs, the JBI quasi-experimental checklist for non-randomized experimental studies is used here as the closest JBI framework. The judgements are therefore a structured adaptation to the experimental animal/in-vitro context, not a claim that these papers are human quasi-experimental studies.

2. Articles and study-design identification

Article	Study design	Experimental model	JBI tool used
Blauwblomme et al. (2018), "Transient ischemia facilitates neuronal chloride accumulation and severity of seizures."	Non-randomized experimental laboratory study with repeated-measures in-vitro experiments.	Organotypic mouse hippocampal slice cultures; oxygen-glucose deprivation (OGD), reperfusion, electrophysiology, two-photon imaging and pharmacological interventions.	JBI Checklist for Quasi-Experimental Studies (adapted to animal/in-vitro experiments).
Silayeva et al. (2015), "KCC2 activity is critical in limiting the onset and severity of status epilepticus."	Non-randomized experimental animal study with complementary in-vitro mechanistic experiments.	WT and KCC2-S940A knock-in mice; kainate-induced status epilepticus; EEG, biochemical assays, acute slices and cultured hippocampal neurons.	JBI Checklist for Quasi-Experimental Studies (adapted to animal/in-vitro experiments).

Design identification is based on the methods and experimental comparisons reported in the papers. The first article manipulates OGD/reperfusion and pharmacological conditions in hippocampal slice cultures; the second manipulates genotype and experimental seizure conditions in mice and cultured neurons.

3. Article 1 – Blauwblomme et al. (2018)

Title: Transient ischemia facilitates neuronal chloride accumulation and severity of seizures

Journal: Annals of Clinical and Translational Neurology, 2018; DOI: 10.1002/acn3.617

Design: Non-randomized experimental laboratory study using organotypic mouse hippocampal slice cultures.

JB1 quasi-experimental checklist

JB1 item	Judgement	Rationale	Article evidence
1. Is it clear what is the cause and what is the effect?	Yes	The experimental manipulation (OGD/reperfusion or drug exposure) precedes the measured seizure/chloride outcomes.	Methods explicitly define the OGD exposure and outcomes; results compare control, OGD and recovery.
2. Were the experimental units included in comparisons similar?	Yes / Unclear	Slice cultures were prepared under standardized conditions, but the paper does not describe random allocation of slices to conditions. Therefore similarity is supported by the common preparation but allocation bias cannot be excluded.	Same mouse hippocampal slice-culture preparation and standardized culture/recording conditions were used.
3. Were the experimental units receiving similar treatment other than the intervention/exposure of interest?	Yes	Experimental conditions were largely standardized, with the main differences being OGD, reperfusion or drug exposure.	ACSF composition, temperature, flow and recording procedures were standardized; pharmacological conditions were specified.
4. Was there a control group?	Yes	Control conditions were explicitly used throughout the experiments.	Control recordings were included; e.g., N=6 slices in the main seizure experiment.
5. Were multiple measurements made before and after the intervention/exposure?	Yes	The design includes repeated measurements during control, OGD and reperfusion; for example, chloride was recorded during a 20-min control period, 30-min OGD and 70-min reperfusion.	120-min imaging experiments included control, OGD and reperfusion periods.
6. Was follow-up/retention complete, and if not, were differences adequately described and analyzed?	Not applicable	Classical longitudinal participant follow-up is not applicable to ex-vivo slice experiments. The reported experiments did not describe attrition as a participant-retention problem.	Ex-vivo experimental design; no conventional participant follow-up.
7. Were outcomes measured in the same way across comparisons?	Yes	Electrophysiological and imaging outcomes were measured using the same stated procedures across conditions.	Field potentials and imaging were obtained with the same instruments/procedures for comparison groups.
8. Were outcomes measured in a reliable way?	Yes	Electrophysiology used extracellular field recordings and defined IED/ILD criteria; chloride was measured with the ratiometric Clomeleon method and two-photon imaging.	ILD/IED definitions, sampling rate, filtering and Clomeleon measurement procedures were described.
9. Was appropriate statistical analysis used?	Yes, with a limitation	Appropriate repeated-measures and nonparametric procedures were reported. A limitation is that cell-level observations are nested within slices, so independence/pseudoreplication should be considered when	Student's t test, repeated-measures ANOVA, Friedman RM ANOVA and Wilcoxon signed-rank tests were reported.

		interpreting some imaging analyses.	
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Overall risk of bias – Article 1

Overall judgement: MODERATE RISK OF BIAS.

The study has a clear experimental sequence, appropriate control conditions, repeated measurements, standardized outcome measurement, and generally appropriate statistical procedures. The main concerns are the absence of a clearly reported random allocation process and the possibility of non-independence/pseudoreplication when many cells are analyzed within a smaller number of slices. Because the paper is an ex-vivo mechanistic experiment, conventional follow-up/retention items are not directly applicable.

4. Article 2 – Silayeva et al. (2015)

Title: KCC2 activity is critical in limiting the onset and severity of status epilepticus

Journal: Proceedings of the National Academy of Sciences (PNAS), 2015; 112(11):3523–3528; DOI: 10.1073/pnas.1415126112

Design: Non-randomized experimental animal study with complementary in-vitro mechanistic experiments.

JB1 quasi-experimental checklist

JB1 item	Judgement	Rationale	Article evidence
1. Is it clear what is the cause and what is the effect?	Yes	The mutation/genotype or experimental treatment precedes measurement of seizure, EEG, biochemical and cellular outcomes.	Genotype and kainate exposure precede measurement of SE onset, lethality, EEG power and KCC2-related outcomes.
2. Were the experimental units included in comparisons similar?	Yes	WT and S940A mice were compared, and the paper specifically reports WT littermates in key experiments. Baseline KCC2 expression/activity and several synaptic markers were comparable.	The paper reports S940A homozygotes and WT littermates with comparable basal KCC2 expression/activity and several synaptic markers.
3. Were the experimental units receiving similar treatment other than the intervention/exposure of interest?	Yes	The experimental conditions were substantially standardized; kainate dose, electrophysiology and biochemical procedures were specified.	Kainate dose and experimental procedures were specified; WT and mutant groups underwent comparable procedures.
4. Was there a control group?	Yes	Vehicle/WT controls and WT littermate comparisons were used.	WT littermates, vehicle-treated animals and untreated/control cellular conditions were used.
5. Were multiple measurements made before and after the intervention/exposure?	Yes / partly	Several experiments used repeated measurements over time, especially electrophysiological/cellular recovery experiments. However, the main in-vivo seizure outcome is a post-kainate comparison rather than a classic pre/post controlled intervention.	Cellular experiments measured baseline, challenge and recovery values; however, the central in-vivo SE comparison is not a classic repeated pre/post design.
6. Was follow-up/retention complete, and if not, were differences adequately described and analyzed?	Yes	Loss/retention problems were not presented as a major issue; the paper reports the experimental numbers for the key comparisons.	Key in-vivo comparisons report n values (e.g., n=6 per genotype for the SE lethality/EEG experiment); no major loss-to-follow-up issue is described.
7. Were outcomes measured in the same way across comparisons?	Yes	WT and S940A groups were evaluated with the same stated assays and experimental procedures.	EEG and biochemical/cellular assays were applied to the comparison groups using the same procedures.
8. Were outcomes measured in a reliable way?	Yes	EEG, perforated patch clamp, immunoblotting, biotinylation and behavioral assays were described, with mechanistic controls such as KCC2 inhibition and AP5.	The paper specifies EEG recording, perforated patch clamp, immunoblotting/biotinylation, behavioral assays and pharmacological controls.
9. Was appropriate statistical analysis used?	Yes	Statistical tests were reported for the comparisons, including	Statistical significance and sample sizes are reported throughout; several

		paired/unpaired t tests and other inferential analyses. Sample sizes were small in several experiments, which reduces precision.	experiments use small n, e.g. n=4–6 for some biochemical/in-vivo comparisons.
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Overall risk of bias – Article 2

Overall judgement: MODERATE RISK OF BIAS.

The paper provides strong experimental controls, uses WT littermates, verifies the baseline comparability of genotypes, and triangulates the proposed mechanism using in-vivo, acute-slice and cultured-neuron experiments. However, allocation/randomization and blinding procedures are not clearly reported, and several experiments have small sample sizes. The JBI quasi-experimental item concerning multiple pre/post outcome measurements is only partly applicable to the central in-vivo SE experiment.

5. Summary of risk-of-bias judgements

Article	Study design	Overall JBI-based judgement	Main concerns
Blauwblomme et al. (2018)	Non-randomized experimental in-vitro study	Moderate risk of bias	No clearly reported random allocation; possible nested-data/pseudoreplication concern; some JBI retention items not applicable.
Silayeva et al. (2015)	Non-randomized experimental animal study	Moderate risk of bias	Randomization/blinding not clearly reported; several small-n experiments; pre/post criterion only partly applicable to the main in-vivo outcome.

6. Final conclusion

Using the JBI quasi-experimental critical appraisal framework as an adapted tool for these laboratory animal/in-vitro studies, both articles are judged to have a moderate overall risk of bias. This judgement does not mean that the studies are invalid or that their findings are false. It indicates that some design/reporting limitations—particularly unclear allocation/blinding procedures and, in the first paper, possible dependence among cell-level observations—should be considered when interpreting the evidence and incorporating these studies into a systematic review.

7. Key methodological sources

1. Joanna Briggs Institute (JBI). Critical Appraisal Tools. JBI Global. <https://jbi.global/critical-appraisal-tools>
2. JBI Critical Appraisal Checklist for Quasi-Experimental Studies (non-randomized experimental studies).
3. Barker TH, Habibi N, Aromataris E, Stone JC, Leonardi-Bee J, Sears K, et al. The revised JBI critical appraisal tool for the assessment of risk of bias for quasi-experimental studies. *JBI Evidence Synthesis*. 2024;22(3):378–388.

8. Article sources

Blauwblomme T, Dzhalal V, Staley KJ, et al. Transient ischemia facilitates neuronal chloride accumulation and severity of seizures. *Ann Clin Transl Neurol*. 2018;5(9):1048–1061.
<https://onlinelibrary.wiley.com/doi/10.1002/acn3.617>

Silayeva L, Deeb TZ, Hines RM, et al. KCC2 activity is critical in limiting the onset and severity of status epilepticus. *Proc Natl Acad Sci USA*. 2015;112(11):3523–3528.
<https://www.pnas.org/doi/full/10.1073/pnas.1415126112>